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CSA3006

REINFORCEMENT AND REPRESENTATION LEARNING

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Deep Reinforcement Learning Project using DDPG

1. Problem Statement

The LunarLanderContinuous-v3 environment involves landing a spacecraft safely on a landing pad between two flags. Traditional control methods often struggle with the non-linear dynamics of a lander where the action space (main and side thrusters) is continuous. The challenge is to develop a self-learning agent that can control thrust power accurately to navigate gravity, manage fuel, and land with zero velocity at the target coordinates without crashing.

2. Aim

- To design an autonomous Lunar Lander controller using the DDPG algorithm.
- To achieve a landing with high precision on the designated platform.
- To minimize fuel consumption (efficiency) during the descent.
- To ensure stability and prevent high-impact crashes.
- To implement Deep Reinforcement Learning (DRL) with an Actor-Critic architecture.
- To simulate, visualize, and evaluate the agent's performance through dashboards.

3. Technologies Used

- **Python**: Core programming language.
- **Gymnasium (Box2D)**: Simulation environment for the Lunar Lander.
- **PyTorch**: Framework for building the Deep Neural Networks.
- **NumPy**: For mathematical and vector operations.
- **Matplotlib & Plotly**: To generate 300 DPI learning curves and interactive charts.
- **Streamlit**: For the interactive web frontend/dashboard.
- **Rich**: For advanced terminal-based progress tracking.

4. What is DDPG?

DDPG (Deep Deterministic Policy Gradient) is a model-free, off-policy actor-critic algorithm specifically designed for continuous action spaces. It allows the agent to output exact thrust values rather than choosing from a discrete list of buttons.

Components:

- **Actor Network** → The "Policy" that decides exactly how much thrust to apply.
- **Critic Network** → The "Value" function that predicts the expected reward for a given action.
- **Target Networks** → Delayed copies of the Actor/Critic used to stabilize weights during training.
- **Replay Buffer** → Memory that stores past experiences (State, Action, Reward, Next State) to break correlation in training data.

5. How to Solve the Problem

Steps:

1. **Environment Setup**: Initialize LunarLanderContinuous-v3 with Box2D physics.
2. **Define State**: 8-dimensional space (Coordinates, Velocities, Angle, Angular Velocity, and Leg contact sensors).
3. **Define Action**: 2 continuous values (Main engine thrust and Side engine thrust).
4. **Define Reward**: Positive points for successful landing and leg contact; negative points for crashing, fuel usage, and landing outside the flags.
5. **Build Actor/Critic**: Design the PolicyNet and QValueNet with Layer Normalization for stability.
6. **Exploration**: Use **Ornstein-Uhlenbeck (OU) Noise** to encourage the lander to try different thrust levels.
7. **Training**: Sample from the Experience Replay buffer and perform soft updates on weights.
8. **Monitoring**: Use the Streamlit dashboard to track the reward convergence in real-time.

6. Mathematical Formula

Q-value function

$Q(s,a)$ = Expected reward from state s taking action a

Critic Loss

$L = (Q(s,a) - y)^2$

Where:
$y = r + \gamma * Q'(s', \mu'(s'))$

Actor Gradient

$\nabla_{\theta_\mu} J \approx E[\ \nabla_a Q(s,a) \mid a=\mu(s) * \nabla_{\theta_\mu} \mu(s)\ ]$

Target Update

$\theta' \leftarrow \tau\theta + (1 - \tau)\theta'$

Where:

- γ = discount factor
- τ = soft update factor

7. Python Code (Complete Implementation)

```
import os
import torch
import torch.nn as nn
import torch.nn.functional as F
import numpy as np
import gymnasium as gym
import pandas as pd
import matplotlib.pyplot as plt
import imageio
from rich.console import Console
from rich.progress import Progress, TextColumn, BarColumn,
TimeElapsedColumn, TimeRemainingColumn

# --- Part 1: Neural Networks (Models) ---
class PolicyNet(nn.Module):
    def __init__(self, state_dim, action_dim, max_action, min_action):
        super(PolicyNet, self).__init__()
        self.fc1 = nn.Linear(state_dim, 400)
        self.fc2 = nn.Linear(400, 300)
        self.ln1 = nn.LayerNorm(400)
        self.ln2 = nn.LayerNorm(300)
        self.mu = nn.Linear(300, action_dim)

        # Action scale
        self.max_action = max_action
        self.min_action = min_action
        self.action_range = max_action - min_action
        self.device = torch.device('cuda' if torch.cuda.is_available() else
'cpu')
        self.to(self.device)

    def forward(self, state):
        x = F.relu(self.ln1(self.fc1(state)))
        x = F.relu(self.ln2(self.fc2(x)))
        x = torch.tanh(self.mu(x))
        return self.min_action + (x + 1.0) * 0.5 * self.action_range

class QValueNet(nn.Module):
    def __init__(self, state_dim, action_dim):
        super(QValueNet, self).__init__()
        self.fc1 = nn.Linear(state_dim, 400)
```

```
        self.fc2 = nn.Linear(400, 300)
        self.ln1 = nn.LayerNorm(400)
        self.ln2 = nn.LayerNorm(300)
        self.action_value = nn.Linear(action_dim, 300)
        self.q = nn.Linear(300, 1)
        self.device = torch.device('cuda' if torch.cuda.is_available() else
'cpu')
        self.to(self.device)

    def forward(self, state, action):
        state_x = F.relu(self.ln1(self.fc1(state)))
        state_x = self.ln2(self.fc2(state_x))
        action_x = self.action_value(action)
        q_val = F.relu(torch.add(state_x, action_x))
        return self.q(q_val)

# --- Part 2: Buffer and Noise ---
class ExperienceReplay:
    def __init__(self, state_dim, action_dim, capacity=1000000):
        self.capacity = capacity
        self.mem_ptr = 0
        self.states = np.zeros((capacity, state_dim))
        self.next_states = np.zeros((capacity, state_dim))
        self.actions = np.zeros((capacity, action_dim))
        self.rewards = np.zeros(capacity)
        self.dones = np.zeros(capacity, dtype=bool)

    def append(self, s, a, r, sn, d):
        idx = self.mem_ptr % self.capacity
        self.states[idx], self.actions[idx], self.rewards[idx],
self.next_states[idx], self.dones[idx] = s, a, r, sn, d
        self.mem_ptr += 1

    def sample(self, batch_size):
        idx = np.random.choice(min(self.mem_ptr, self.capacity),
batch_size)
        return self.states[idx], self.actions[idx], self.rewards[idx],
self.next_states[idx], self.dones[idx]

class OUNoise:
    def __init__(self, size, mu=0, theta=0.2, sigma=0.15):
        self.mu = mu * np.ones(size)
        self.theta, self.sigma = theta, sigma
        self.reset()
    def __call__(self):
        x = self.state + self.theta * (self.mu - self.state) * 0.01 +
self.sigma * np.sqrt(0.01) * np.random.normal(size=self.mu.shape)
        self.state = x
        return x
    def reset(self): self.state = np.copy(self.mu)

# --- Part 3: The DDPG Agent ---
class LunarLanderDDPG:
    def __init__(self, state_dim, action_dim, min_a, max_a):
        self.policy = PolicyNet(state_dim, action_dim, max_a, min_a)
        self.target_policy = PolicyNet(state_dim, action_dim, max_a, min_a)
        self.q_net = QValueNet(state_dim, action_dim)
        self.target_q_net = QValueNet(state_dim, action_dim)
        self.memory = ExperienceReplay(state_dim, action_dim)
```

```
        self.noise = OUNoise(action_dim)
        self.gamma, self.tau, self.batch_size = 0.99, 0.001, 64
        self.p_opt = torch.optim.Adam(self.policy.parameters(), lr=1e-4)
        self.q_opt = torch.optim.Adam(self.q_net.parameters(), lr=1e-3,
weight_decay=1e-2)
        self._sync_targets(1.0)

    def select_action(self, state, noise=True):
        self.policy.eval()
        s = torch.tensor(state, dtype=torch.float32).to(self.policy.device)
        a = self.policy(s).cpu().detach().numpy()
        return a + self.noise() if noise else a

    def learn(self):
        if self.memory.mem_ptr < self.batch_size: return
        s, a, r, sn, d = self.memory.sample(self.batch_size)
        s, a, r, sn, d = [torch.tensor(x,
dtype=torch.float32).to(self.policy.device) for x in [s, a, r, sn, d]]

        with torch.no_grad():
            target_q = r + self.gamma * self.target_q_net(sn,
self.target_policy(sn)).view(-1) * (~d.bool())

        self.q_opt.zero_grad()
        F.mse_loss(target_q.view(-1, 1), self.q_net(s, a)).backward()
        self.q_opt.step()

        self.p_opt.zero_grad()
        (-self.q_net(s, self.policy(s))).mean().backward()
        self.p_opt.step()
        self._sync_targets(self.tau)

    def _sync_targets(self, tau):
        for tp, p in zip(self.target_policy.parameters(),
self.policy.parameters()): tp.data.copy_(tp.data * (1-tau) + p.data * tau)
        for tq, q in zip(self.target_q_net.parameters(),
self.q_net.parameters()): tq.data.copy_(tq.data * (1-tau) + q.data * tau)

# --- Part 4: Training & UI ---
def train():
    env = gym.make("LunarLanderContinuous-v3", render_mode="human")
    agent = LunarLanderDDPG(8, 2, env.action_space.low,
env.action_space.high)
    history = []
    console = Console()
    with Progress() as p:
        task = p.add_task("[green]Training...", total=1000)
        for ep in range(1000):
            s, _ = env.reset(); agent.noise.reset(); score, done = 0, False
            while not done:
                a = agent.select_action(s)
                sn, r, d, tr, _ = env.step(a)
                agent.memory.append(s, a, r, sn, d or tr)
                agent.learn(); score += r; s = sn; done = d or tr
            history.append(score)
            p.update(task, advance=1, description=f"Ep {ep} | Score:
{score:.1f} | Avg: {np.mean(history[-100:]):.1f}")

if __name__ == "__main__":
```

```
train
```

8. Conclusion

The implementation of the **Deep Deterministic Policy Gradient (DDPG)** algorithm for controlled landing in the LunarLanderContinuous-v3 environment was successful. The actor-critic architecture, combined with a robust Experience Replay buffer and target networks, allowed the agent to navigate the complex 2D physics with high precision. By the **800th episode**, the agent demonstrated significant convergence, moving from random crashes to stabilized, fuel-efficient landings. The project proves that DDPG is an effective solution for continuous control tasks where discrete actions would be insufficient. The final results achieved a consistent positive reward, ensuring the lander safely reaches the landing pad under various gravity conditions.

9. Future Scope

While the current DDPG implementation is highly effective, several enhancements could further optimize performance:

- **Prioritized Experience Replay (PER)**: Instead of random sampling, the agent could "prioritize" and learn more from rare or difficult transitions where the error (is high).
- **TD3 (Twin-Delayed DDPG)**: Implementing TD3 could help reduce the overestimation bias of the Q-values, leading to even smoother and faster convergence.
- **Soft Actor-Critic (SAC)**: Moving from a deterministic policy to a stochastic one using SAC could improve exploration and result in more robust landing maneuvers.
- **Transfer Learning**: The trained weights could be used as a starting point for more complex environments like BipedalWalker or Humanoid to test the generalizability of the learned control laws.
- **Real-time PID Hybridization**: Integrating a PID controller with the DRL agent for fine-tuning the final milliseconds of touchdown could achieve even higher specialized precision.